

Visual basic 6.0

(Long question solved)

Q1: What is an Array? Explain 1D and 2D Array. Write a program that gets five input from the user in an array and then sorts this array in ascending order.

Answer:

An array in Visual Basic is a special variable that can hold more than one value at a time. It is a data structure that stores elements of the same data type, such as integers or strings, under one variable name but at different index positions.

➤ 1D Array (One-Dimensional Array):

It stores data in a single row, like a list.

Example: Dim numbers(4) As Integer This can store 5 values (0 to 4).



➤ 2D Array (Two-Dimensional Array):

It stores data in a grid of rows and columns.

Example: Dim matrix(2, 2) As Integer This creates a 3x3 matrix (from index 0 to 2).

➤ Benefits of Arrays:

Efficient storage of multiple values

Easy looping and sorting

Useful for mathematical and data-related tasks

❓ Program to take 5 values from user and sort in ascending order:



Dim arr(4) As Integer


```
Dim i As Integer, j As Integer, temp As Integer
```

```
' Input values
```

```
For i = 0 To 4
```

```
arr(i) =
```

```
Val(InputBox("Enter number " & (i + 1)))
```

```
Next
```

```
' Sorting using Bubble Sort
```

```
For i = 0 To 3
```

```
For j = i + 1 To 4
```

```
If arr(i) > arr(j) Then
```

```
temp = arr(i)
```

```
arr(i) = arr(j)
```

```
arr(j) = temp
```

```
End If
```

```
Next
```

```
Next
```

```
' Display sorted array
```

```
For i = 0 To 4
```



MessageBox "Sorted number " & (i + 1) & ": " & arr(i)

Next

Easy Explanation:

Array ek aisi variable hoti hai jo ek naam mein bohat sari values store kar sakti hai. Agar humein 5 numbers store karne hon to hum 5 alag variables na bana kar sirf ek array banate hain jaise arr(4).

1D array aik line mein data rakhti hai jaise list.

2D array rows aur columns ka system hota hai jaise matrix.

Upar diye gaye program mein:

Humne user se 5 numbers liye.

Phir unko Bubble Sort algorithm se ascending order mein sort kiya.



Aakhir mein MessageBox se har number ko show kiya.

✓ **Q2: Explain Nested-If statements. Write a program that inputs three numbers and display the smallest number using nested-if.**

Answer:

A nested-if statement means placing one If...Then...Else statement inside another. It is used to make more detailed decisions when multiple conditions must be checked.

➤ Syntax:

If condition1 Then If condition2 Then ' Do something Else ' Do something else End If Else ' Other condition End If

➤ Usefulness:

Useful when one decision depends on the result of another



Helps in checking multiple layers of conditions

Used in login systems, grading systems, comparisons, etc.

❓ Program to find smallest of three numbers:

```
Dim a As Integer,
```

```
b As Integer,
```

```
c As Integer
```

```
a = Val(InputBox("Enter first number"))
```

```
b = Val(InputBox("Enter second number"))
```

```
c = Val(InputBox("Enter third number"))
```

```
If a < b Then If a < c
```

```
Then
```

```
MsgBox "Smallest number is: " & a
```

```
Else
```

```
MsgBox "Smallest number is: " & c
```

```
End If
```

```
Else
```



If $b < c$ Then

MsgBox "Smallest number is: " & b

Else

MsgBox "Smallest number is: " & c

End If

End If

Easy Explanation:

Nested-if ka matlab hota hai aik if ke andar doosra if. Ye tab use hota hai jab aik condition ke andar aur condition check karni ho.

Upar wale program mein:

Teen numbers user se liye gaye.

Nested-if ke zariye pehle check kiya ke $a < b$ hai ya nahi.



Phir $a < c$ ya $b < c$ ko check karke choti value print kar di.

✓ **Q3: Explain the steps required to create a menu on a form using the menu editor.**

Answer:

To create a menu on a form using the Menu Editor in Visual Basic 6.0, follow these steps:

➤ **Steps:**

Open the Form:

Open your form in the VB6 IDE.

Open Menu Editor:

Go to Tools → Menu Editor

Or press Ctrl + E

Add Menu Items:

In the Caption field, type the text you want on the menu bar
(like File)

In the Name field, give it an identifier (like mnuFile)

Click Next to add more items.

Add Sub-Menus:

Click on the Right Arrow button to make it a submenu (like Open, Save, Exit under File)

Set their names like mnuOpen, mnuSave, etc.

Set Shortcut Keys (Optional):

You can set shortcut keys like Ctrl + S, Ctrl + O for quick access.

Code the Menu Items:

Double-click on any menu item and write VB code under its click event.

Example: Private Sub mnu

Q4: Write a program that takes a number from user, prints its Factorial and also prints its Table.

Answer:



This question has two parts:

- Factorial: The factorial of a number n (written as $n!$) is the product of all positive integers from 1 to n . Example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- Table: The multiplication table of the number, usually from 1 to 10.

VB Code:

```
Dim num As Integer
```

```
Dim factorial As Long
```

```
Dim i As Integer
```

```
num = Val(InputBox("Enter a number:"))
```

```
factorial = 1
```

```
' Calculate Factorial
```

```
For i = 1 To num
```

```
factorial = factorial * i
```

```
Next
```




```
MsgBox "Factorial of " & num & " is " & factorial
```

```
' Print Table
```

```
Dim table As String
```

```
table = "Table of " & num & ":" & vbCrLf
```

```
For i = 1 To 10
```

```
table = table & num &
```

```
" x " & i & " = " & num * i & vbCrLf
```

```
Next
```

```
MsgBox table
```

Easy Explanation:

Is program mein user se aik number input liya jata hai. Phir uska factorial calculate kiya jata hai using loop. Factorial ka matlab hota hai: $\text{number} \times (\text{number}-1) \times \dots \times 1$.

Phir usi number ka 1 se 10 tak ka table print kiya jata hai using loop aur MsgBox ke zariye dikhaya jata hai.



✓ Q5: Write a program to find the eligibility of admission for a professional course.

Criteria:

Marks in Math ≥ 65

Marks in Physics ≥ 55

Marks in Chemistry ≥ 50

VB Code:

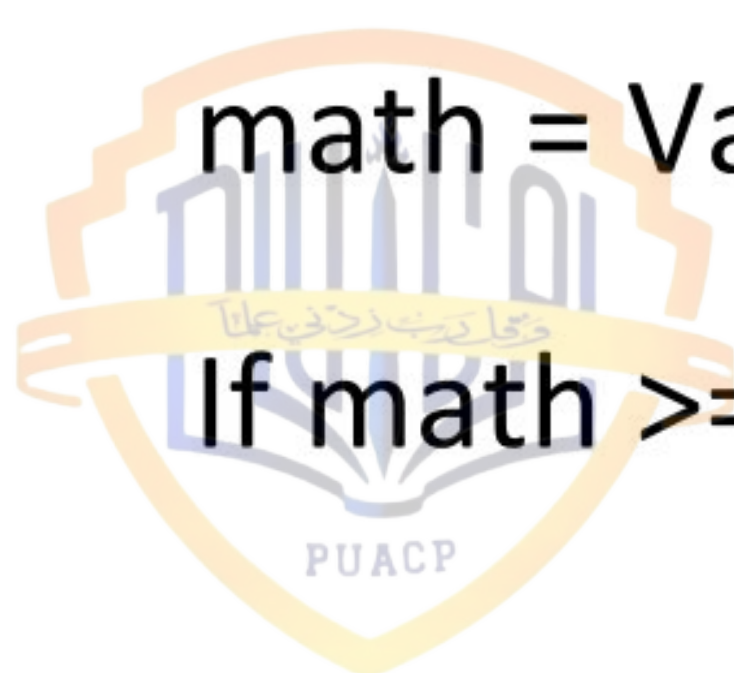
```
Dim phy, chem, math As Integer
```

```
phy = Val(InputBox("Enter marks in Physics:"))
```

```
chem = Val(InputBox("Enter marks in Chemistry:"))
```

```
math = Val(InputBox("Enter marks in Mathematics:"))
```

```
If math  $\geq 65$  And phy  $\geq 55$  And chem  $\geq 50$  Then
```



MsgBox "The candidate is eligible for the course."

Else

MsgBox "The candidate is not eligible for the course."

End If

Easy Explanation:

Is program mein teen subjects ke marks input liye jate hain:
Physics, Chemistry, aur Maths. Eligibility condition yeh hai:

Maths mein 65 ya zyada

Physics mein 55 ya zyada

Chemistry mein 50 ya zyada

Agar ye teeno shartein puri hoti hain, toh candidate eligible
hota hai. Warna nahi.



✓ Q6: Write a program that takes 5 values in an array from user. Calculates and print Sum and Average on the screen.

VB Code:

```
Dim arr(4) As Integer
```

```
Dim i, sum As Integer
```

```
Dim average As Single
```

```
sum = 0
```

```
For i = 0 To 4
```

```
arr(i) =
```

```
Val(InputBox("Enter number " & (i + 1) & ":")) sum = sum +  
arr(i)
```

```
Next
```

```
average = sum / 5
```

```
MsgBox "Sum = " & sum & vbCrLf & "Average = " & average
```



Easy Explanation:

Yeh program 5 numbers user se input leta hai aur unka total (sum) aur average calculate karta hai.

Loop ke zariye array mein values store hoti hain.

Sum ko calculate karne ke baad usay 5 se divide karke average nikala jata hai.

